From Vapnik to Kernel SVM
A Pedagogical Journey from Hyperplanes to Kernels

Mathematical origins, key theorems, and historical anecdotes

Prepared for an Advanced Machine Learning course

Latentia Labs / Teaching support

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Learning objectives

By the end of this mini-lecture, students should be able to

- explain why SVMs start from **linear algebra**: vectors, dot products, norms, projections;
- derive the geometric meaning of the **maximum margin hyperplane**;
- understand why the dual formulation depends only on **dot products**;
- explain the **kernel trick** as an implicit feature-space construction;
- connect SVMs to earlier mathematical ideas: Mercer kernels, RKHS, and potential functions.

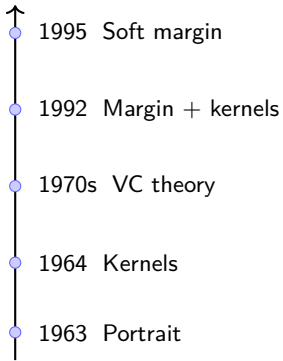
Core message

A kernel SVM is not "magic nonlinearity". It is a **linear separator in a richer space**, computed without explicitly visiting that space.

The historical thread: from portraits to kernels

A compact historical line

- **1963**: Vapnik and Lerner - generalized portrait method.
- **1964**: Aizerman, Braverman, Rozonoer - potential functions, early kernel-like idea.
- **1970s**: Vapnik and Chervonenkis - statistical learning theory and capacity control.
- **1992**: Boser, Guyon, Vapnik - optimal margin classifiers with kernels.
- **1995**: Cortes and Vapnik - soft-margin support-vector networks.



Sources: Vapnik/Lerner 1963; Aizerman et al. 1964; Boser/Guyon/Vapnik 1992; Cortes/Vapnik 1995.

Anecdote: the SVM is a fusion, not a single trick

The misleading story

"Someone invented a clever nonlinear classifier called SVM."

The better historical story

The modern SVM is the meeting point of three older rivers:

1. **Geometry**: separate classes by a hyperplane.
2. **Statistical learning**: control model capacity to generalize.
3. **Functional analysis**: use kernels to compute inner products in hidden spaces.

Teaching punchline

The kernel trick becomes obvious only after the SVM is written in its **dual form**. The algebra reveals the trick.

Step 1 - Classification as separation

Data

Training set:

$$(x_i, y_i)_{i=1}^n, \qquad x_i \in \mathbb{R}^d, \quad y_i \in \{-1, +1\}.$$

We seek a decision rule:

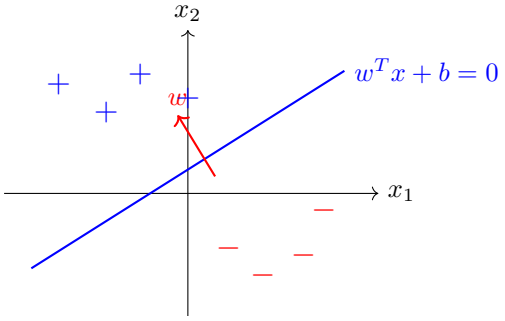
$$f(x) = \text{sign}(w^T x + b).$$

Linear algebra object

The equation

$$w^T x + b = 0$$

defines an affine hyperplane. The vector w is normal to that hyperplane.



Theorem 1 - Distance from a point to a hyperplane

Theorem

For the hyperplane

$$H = \{x \in \mathbb{R}^d : w^T x + b = 0\}, \qquad w \neq 0,$$

the signed distance from a point x to H is

$$\frac{w^T x + b}{\|w\|},$$

and the Euclidean distance is

$$\boxed{\operatorname{dist}(x, H) = \frac{|w^T x + b|}{\|w\|}.}$$

Proof idea

Project x onto the normal direction $w/\|w\|$. Only the component of x along w changes the value of $w^T x + b$.

Step 2 - The margin idea

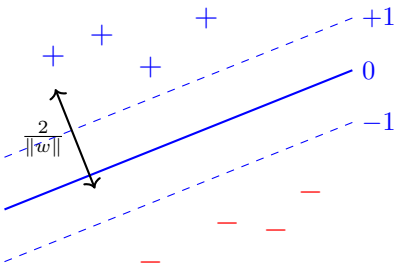
Canonical scaling

If the data are separable, we can rescale (w, b) so that

$$y_i(w^T x_i + b) \geq 1.$$

Then the two margin hyperplanes are

$$w^T x + b = +1, \qquad w^T x + b = -1.$$



Vapnik's geometric principle

Do not choose just any separator. Choose the separator that leaves the largest empty corridor between classes.

Proposition 1 - Width of the margin

Proposition

The distance between the parallel hyperplanes

$$w^T x + b = +1 \quad \text{and} \quad w^T x + b = -1$$

is

$$\boxed{\frac{2}{\|w\|}.}$$

Proof

Take one point x_+ on $w^T x + b = +1$ and move along the normal direction to reach the plane $w^T x + b = -1$:

$$x_- = x_+ - \lambda \frac{w}{\|w\|}.$$

Plug into the equation:

$$w^T x_- + b = 1 - \lambda \|w\| = -1.$$

Therefore $\lambda = 2/\|w\|$.

Step 3 - Maximum margin becomes convex optimization

Hard-margin SVM primal problem

Maximizing $2/\|w\|$ is equivalent to minimizing $\frac{1}{2}\|w\|^2$:

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2}\|w\|^2 \\ \text{s.t.} \quad & y_i(w^T x_i + b) \geq 1, \quad i = 1, \ldots, n. \end{aligned}$$

Why this mattered historically

The classifier is no longer a heuristic. It is the solution of a **convex quadratic program**: one global optimum, not many local traps.

Step 4 - Lagrangian and KKT conditions

Lagrangian

Introduce multipliers $\alpha_i \geq 0$:

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^{n} \alpha_i \left[y_i (w^T x_i + b) - 1 \right].$$

Stationarity

$$\frac{\partial \mathcal{L}}{\partial w} = 0 \quad \Rightarrow \quad \boxed{w = \sum_{i=1}^{n} \alpha_i y_i x_i}$$

$$\frac{\partial \mathcal{L}}{\partial b} = 0 \quad \Rightarrow \quad \boxed{\sum_{i=1}^{n} \alpha_i y_i = 0.}$$

A first surprise

The normal vector w is a weighted combination of training examples

Theorem 2 - The support-vector representation

Theorem

At the optimum of the hard-margin SVM, the decision function can be written as

$$f(x) = \operatorname{sign}\left(\sum_{i=1}^{n} \alpha_i y_i \langle x_i, x \rangle + b\right),$$

where $\alpha_i \geq 0$ and only points on the margin may have $\alpha_i > 0$.

Why only margin points matter

The KKT complementarity condition gives

$$\alpha_i \left[y_i(w^T x_i + b) - 1\right] = 0.$$

If a point is strictly outside the margin, then $y_i(w^T x_i + b) > 1$, so $\alpha_i = 0$.

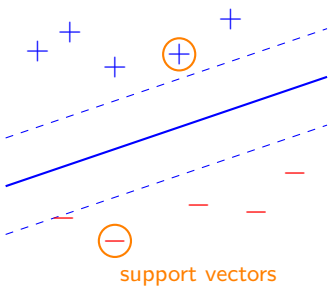
Anecdote: the boundary writes the classifier

Pedagogical anecdote

In an SVM, many training points do not directly appear in the final formula. The classifier is carried by the few points closest to the decision boundary: the **support vectors**.

A memorable sentence

The easy points vote during training; the hard boundary points write the final classifier.



Step 5 - The dual problem reveals the kernel door

Dual hard-margin SVM

$$\max_{\alpha} \quad \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{n} \alpha_i \alpha_j y_i y_j \ \langle x_i, x_j \rangle$$

$$\text{s.t.} \quad \alpha_i \geq 0, \quad \sum_{i=1}^{n} \alpha_i y_i = 0.$$

Key observation

The training data appear only through inner products:

$$\langle x_i, x_j \rangle .$$

This is the exact algebraic place where kernels enter.

Step 6 - A nonlinear problem

In the input plane

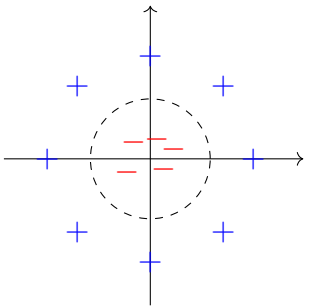
Some data are not linearly separable in $\mathbb{R}^2$. A straight line cannot separate concentric patterns.

Idea

Map the data into a higher-dimensional space:

$$\phi : \mathbb{R}^d \to \mathcal{H}.$$

Then learn a linear hyperplane in $\mathcal{H}$.



not linearly separable in input space

The kernel trick

Definition

A kernel is a function

$$K(x, z) = \langle \phi(x), \phi(z) \rangle_{\mathcal{H}}$$

for some feature map ϕ into an inner-product space $\mathcal{H}$.

The trick

Replace every dot product in the dual by a kernel:

$$\langle x_i, x_j \rangle \quad \rightsquigarrow \quad K(x_i, x_j).$$

Then the SVM behaves as if it had built a hyperplane in feature space, without explicitly computing $\phi(x)$.

Kernelized decision rule

$$f(x) = \operatorname{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b \right).$$

Proposition 2 - Polynomial kernel example

Proposition

For $x, z \in \mathbb{R}^2$, the degree-2 polynomial kernel

$$K(x, z) = (x^T z)^2$$

is an inner product in a transformed space.

Proof

Let

$$\phi(x) = \begin{pmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \end{pmatrix}.$$

Then

$$\langle \phi(x), \phi(z) \rangle = x_1^2 z_1^2 + 2x_1x_2z_1z_2 + x_2^2 z_2^2 = (x_1z_1 + x_2z_2)^2.$$

Therefore

$$K(x, z) = \langle \phi(x), \phi(z) \rangle .$$

Theorem 3 - Mercer condition, informal version

Mercer-type statement

A continuous symmetric function K can act as a valid kernel when its Gram matrix is positive semidefinite for any finite sample:

$$G_{ij} = K(x_i, x_j), \qquad c^T G c \geq 0 \quad \forall c \in \mathbb{R}^n.$$

Why this matters

Positive semidefiniteness is what allows us to interpret K as an inner product in some feature space:

$$K(x, z) = \langle \phi(x), \phi(z) \rangle_{\mathcal{H}}.$$

Historical note

Mercer studied positive kernels in 1909. Aronszajn later gave the RKHS framework in 1950.

RKHS: the hidden space behind kernels

Definition idea

A reproducing kernel Hilbert space is a Hilbert space of functions where evaluation at a point can be written as an inner product:

$$f(x) = \langle f, K(x, \cdot) \rangle_{\mathcal{H}} .$$

Reproducing property

The kernel reproduces function values:

$$\boxed{f(x) = \langle f, K_x \rangle_{\mathcal{H}} , \qquad K_x(\cdot) = K(x, \cdot).}$$

Connection to SVM

Kernel SVMs can be interpreted as learning a separating function in an RKHS while controlling its norm.

Theorem 4 - Representer intuition

Representer theorem, informal version

For many regularized learning problems in an RKHS,

$$\min_{f \in \mathcal{H}} \sum_{i=1}^{n} L(y_i, f(x_i)) + \lambda \|f\|_{\mathcal{H}}^2 ,$$

the minimizer has the finite expansion

$$\boxed{f^*(x) = \sum_{i=1}^{n} c_i K(x_i, x).}$$

Pedagogical interpretation

Even if the hidden feature space is infinite-dimensional, the learned function is built from a finite number of kernel evaluations against training examples.

Common kernels and their intuition

Kernel	Formula	Intuition
Linear	$x^T z$	original input space
Polynomial	$(\gamma x^T z + r)^p$	interactions up to degree p
RBF/Gaussian	$\exp(-\gamma \|x - z\|^2)$	local similarity bumps
Sigmoid	$\tanh(\gamma x^T z + r)$	neural-network-inspired, not always PSD

Important teaching warning

A kernel is not just a similarity score. For SVM optimization, it must define a valid positive-semidefinite Gram matrix.

Soft margin: when real data are noisy

Problem

Real datasets are rarely perfectly separable. Cortes and Vapnik introduced slack variables to allow controlled violations.

Soft-margin primal

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i \\ \text{s.t.} \quad & y_i(w^T x_i + b) \geq 1 - \xi_i, \\ & \xi_i \geq 0. \end{aligned}$$

Meaning of C

C controls the trade-off between a wide margin and training violations.

Hyperparameters: mathematical meaning

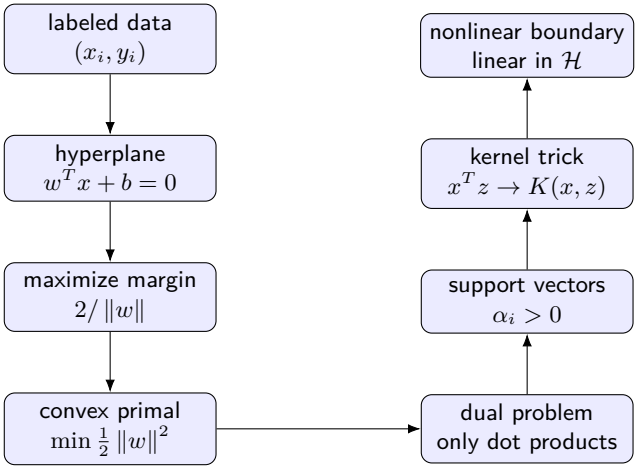
Key parameters

- C: penalty for margin violations. Large C fits training data more aggressively.
- γ in RBF: controls locality. Large γ creates narrow similarity regions.
- degree p in polynomial kernels: controls interaction complexity.

Capacity control viewpoint

SVMs are historically tied to Vapnik's larger program: control the class of functions so that empirical success becomes generalization, not memorization.

One-slide derivation map



Mini-board activity for students

Exercise 1 - Geometry

Given $w = (3, 4)$ and $b = -5$, compute the distance from $x = (1, 2)$ to the hyperplane $w^T x + b = 0$.

Exercise 2 - Margin

For the canonical SVM margin planes $w^T x + b = \pm 1$, prove that the margin width is $2/\|w\|$.

Exercise 3 - Kernel trick

Show explicitly that

$$K(x, z) = (x^T z + 1)^2$$

corresponds to a feature map containing constants, first-order terms, and second-order terms.

Final synthesis

The conceptual chain

linear algebra $\rightarrow$ geometry $\rightarrow$ convex optimization $\rightarrow$ duality $\rightarrow$ kernels $\rightarrow$ nonlinear classification

Take-home message

The kernel SVM is powerful because it combines two ideas that look contradictory:

- it keeps the mathematical discipline of a linear maximum-margin classifier;
- it obtains nonlinear decision boundaries by changing the geometry of the space.

Phrase to remember

The SVM does not chase the most flexible boundary first. It first asks: which boundary can generalize?

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